INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6112

Inactive for new design
after 7 March 1997.

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described
herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Twin triode, subminiature, high Mu.

| | | |
|---|---|---|
| Outline | --- | 3-1 (EIA). |
| Base | --- | E8-10. |
| Envelope | --- | T3. |
| Cathode | --- | Coated unipotential. |

Base connections:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Element | --- | 2a | 2g | h | 2k | 1k | h | 1g | 1a |

ABSOLUTE RATINGS:

| Parameter: | Ef | Eb | Ec | Ehk | Rk/k | Rg/g | Ib/a | Pp/a | TE | Alt |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | v | Ohms | MegΩ | mAdc | W | °C | ft |
| Maximum: | 6.6 | 165 | 0, -55 | 200 | --- | 1.1 | 3.3 | 0.30 | +220 | See Note 1 |
| Minimum: | 6.0 | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| Test Conditions: | 6.3 | 100 | 0 | 0 | 1,500 | --- | --- | --- | --- | --- |

GENERAL:

Qualification – Not required.

First article test is required and shall consist of all tests in table I with a sample size of 2 for a lot size less than or equal to 150 units and a sample size of 4 for a lot size greater than or equal to 151 units. All samples shall pass conformance inspection part 1 of table I before continuing. Half of the samples shall then be subjected to conformance inspection part 2, and the remaining samples shall be subjected to part 3, with no test failures permitted during any testing.

After first article approval, acceptance testing shall consist of conformance inspection part 1 of table I with sample size in accordance with table III, category XVI (acceptance level 10.0) of MIL-PRF-1.

AMSC N/A FSC 5960

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TABLE I. Testing and inspection.

| Requirement or test | MIL-STD-1311 Method | Conditions | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 1 | | | | | | |
| Electrode current (1) (anode) | 1256 | See note 2 and 3 | Ib | 0.5 | 1.10 | mA dc |
| Electrode current (2) (anode) | 1256 | Ec = -2.8 V dc;<br>Rk = 0<br>(see note 2) | Ib | --- | 50 | µA dc |
| Total grid current | 1266 | Eb = 150 V dc;<br>Ec = 0;<br>Rk/k = 820 ohms;<br>Rg/g = 1.0 MegΩ<br>(see notes 2 and 3) | Ic | 0 | -0.3 | µA dc |
| Heater current | 1301 | | If | 280 | 320 | mA |
| Transconductance (1) | 1306 | See note 2 | Sm | 1,500 | 2,100 | µmhos |
| Heater-cathode leakage | 1336 | See note 2 | Ihk | --- | 5.0 | µA dc |
| Short and discontinuity detection | 1201 | | --- | --- | --- | --- |
| Conformance inspection, part 2 | | | | | | |
| Insulation of electrodes | 1211 | See note 2 | --- | 50 | --- | --- |
| Low-frequency vibration | 1031 | Rp = 10,000 ohms;<br>E = 40 Hz; 15 G<br>(see note 2) | Ep | --- | 25 | mV ac |
| Audio frequency noise | 1246 | Esig = 45 mV ac;<br>Rg = 0.5 MegΩ;<br>Rp = 0.2 MegΩ;<br>Rk = 750 ohms;<br>(see note 4) | EB | --- | 17 | vu |
| Grid currents | 1266 | Ef = 7.5 V;<br>Ec = -4.0 V dc;<br>Eb = 150 V dc;<br>Rk = 0;<br>Rg/g = 1.0 MegΩ<br>(see notes 2 and 5) | Ic | 0 | -0.5 | µA dc |
| Transconductance (2) | 1306 | Ef = 5.7 V<br>(see note 2) | $\frac{\Delta Sm}{Ef}$ | --- | 15 | % |
| Amplification factor | 1316 | See note 2 | Mu | 60 | 80 | --- |
| AC amplification | 1321 | Ebb = 100 V dc;<br>Ecc = 0;<br>Esig = 0.2 V ac;<br>Rk = 0<br>(see note 2) | Ep | 8.0 | --- | V ac |



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TABLE I. Testing and inspection - Continued.

| Requirement or test | MIL-STD-1311 Method | Conditions | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 2 - Continued | | | | | | |
| Direct-inter-electrode capacitance | 1331 | No shield (see note 2)<br>No shield (see note 2)<br>No shield; section 1<br>No shield; section 2<br>No shield;<br>No shield | Cgp<br>Cin<br>Cout<br>Cout<br>Cgg<br>Cpp | 0.8<br>1.30<br>0.16<br>0.21<br>---<br>--- | 1.20<br>2.10<br>0.30<br>0.35<br>0.014<br>0.80 | pF<br>pF<br>pF<br>pF<br>pF<br>pF |
| Lead fatigue | 1116 | | --- | --- | --- | --- |
| Envelope strain | 2126 | | --- | --- | --- | --- |
| Shock | 1041 | 450 G;<br>EhK = +100 V dc;<br>Rg = 0.1 MegΩ;<br>(see note 6) | --- | --- | --- | --- |
| Vibration fatigue | 1031 | 2.5 G; fixed frequency;<br>F = 25 min, 60 max | --- | --- | --- | --- |
| Post-shock and vibration-fatigue test end points: | --- | | | | | |
| Low-frequency vibration | 1031 | | Ep | --- | 100 | mV ac |
| Heater-cathode leakage | 1336 | | Ihk | --- | 20 | µA dc |
| Change in transconductance (1) of individual tubes | 1306 | | ΔSmt | --- | 20 | % |
| Permanence of marking | 1105 | | | | | |
| Conformance inspection, part 3 | | | | | | |
| Heater-cycling life | 1506 | Ef = 7.0 V;<br>1 min “on”, 4 min “off”;<br>Ehk = 140 V ac;<br>Ec = Eb = 0 | --- | --- | --- | --- |
| Stability life | 1516 | Eb = 150 V dc;<br>Ehk = +200 V dc;<br>Rg/g = 1.0 MegΩ;<br>Rk/k = 820 ohms;<br>TA = room<br>(see note 2) | --- | --- | --- | --- |
| Stability life-test end point: | --- | | | | | |
| Change in transconductance (1) of individual tubes | 1306 | | ΔSmt | --- | 10 | % |
| Intermittent life (room temperature) | 1501 | Stability life test, or equivalent conditions; TA = room | | | | |
| Intermittent life-test end point (room temperature) (500 hours): | --- | | | | | |
| Insulation of electrodes | 1211 | | R | 50 | --- | MegΩ |



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TABLE I. Testing and inspection - Continued.

| Requirement or test | MIL-STD-1311 Method | Conditions | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Conformance inspection, part 3 - Continued | | | | | | |
| Intermittent life-test end point (room temperature) (1,000 hours): | --- | | | | | |
| Insulation of electrodes | 1211 | | R | 25 | --- | MegΩ |
| Intermittent life (high temperature) | 1501 | Stability life-test conditions; TE = +220°C (min) (see notes 2 and 7) | --- | --- | --- | --- |
| Intermittent life-test end points (500 hours) (high temperature): | --- | | | | | |
| Total grid current | 1266 | | Ic | 0 | -0.9 | μA dc |
| Heater current | 1301 | | If | 276 | 328 | mA |
| Change in transconductance (1) of individual tubes | 1306 | | $\frac{\Delta Sm}{t}$ | --- | 20 | % |
| Transconductance (2) | 1306 | | $\frac{\Delta Sm}{Ef}$ | --- | 15 | % |
| Heater-cathode leakage | 1336 | | Ihk | --- | 10 | μA dc |
| Insulation of electrodes | 1211 | | R | 50 | --- | MegΩ |
| Transconductance (1) average change | 1306 | | Avg $\frac{\Delta Sm}{t}$ | --- | 15 | % |
| Intermittent life-test end points (1,000 hours) (high temperature): | --- | | | | | |
| Total grid current | 1266 | | Ic | 0 | -0.9 | μA dc |
| Heater current | 1301 | | Il | 276 | 328 | mA |
| Change in transconductance (1) of individual tubes | 1306 | | $\frac{\Delta Sm}{t}$ | --- | 25 | % |
| Transconductance (2) | 1306 | | $\frac{\Delta Sm}{Ef}$ | --- | 20 | % |
| Heater-cathode leakage | 1336 | | Ihk | --- | 10 | μA dc |
| Insulation of electrodes | 1211 | | R | 25 | --- | MegΩ |

NOTES:

1. See “Reduced pressure (altitude) rating”, and altitude, maximum peak voltage in the basic document.
2. Test each unit separately.
3. This test to be performed at the conclusion of the holding period.
4. Tie 1k to 2k; 1g to 2g; and 1a to 2a.



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NOTES: - Continued.

5. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

| Ef | Ec1 | Eb | Rk | Rg |
|---|---|---|---|---|
| V | V dc | V dc | Ohms | MegΩ |
| 7.5 | 0 | 150 | 820 | 1.0 |

6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron type short indicator is used.

7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Referenced documents. In addition to MIL-PRF-1 this specification sheet references MIL-STD-1311.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2013-026)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 99

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